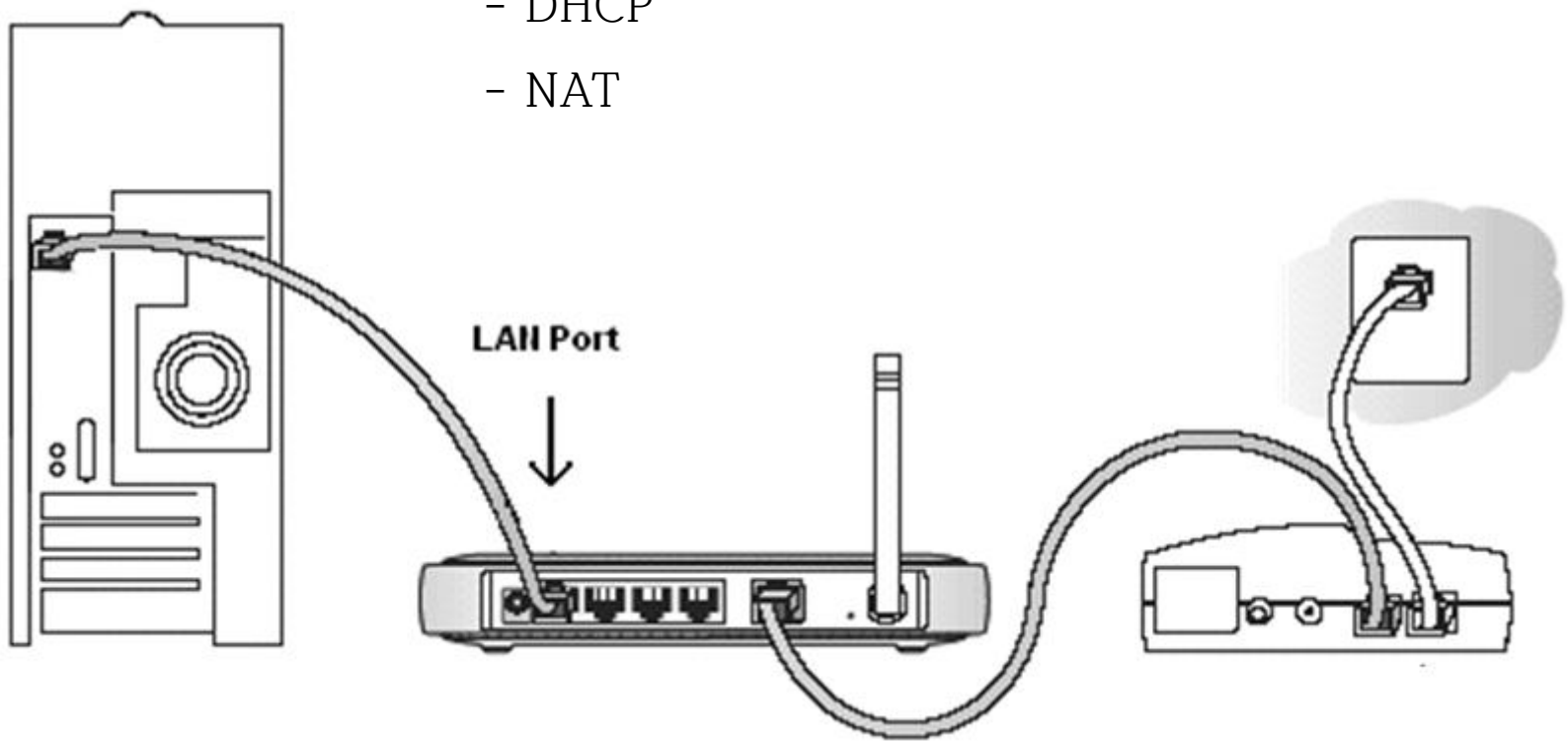
NAT

SoHo(Small office/home office)에서의 NAT

- 공유기 기능
  - 패킷 전달
  - DHCP
  - NAT



❶ 컴퓨터 ❷ 유/무선 공유기 ❸ 모뎀

NAT 기능

- 네트워크 주소 변환 (Network Address Translation)
- 송신지 또는 수신지 IP address를 다른 주소로 변환
- Source NAT(SNAT)와 Destination NAT(DNAT)

송신지 IP	수신지 IP
192.168.1.10	200.10.10.10

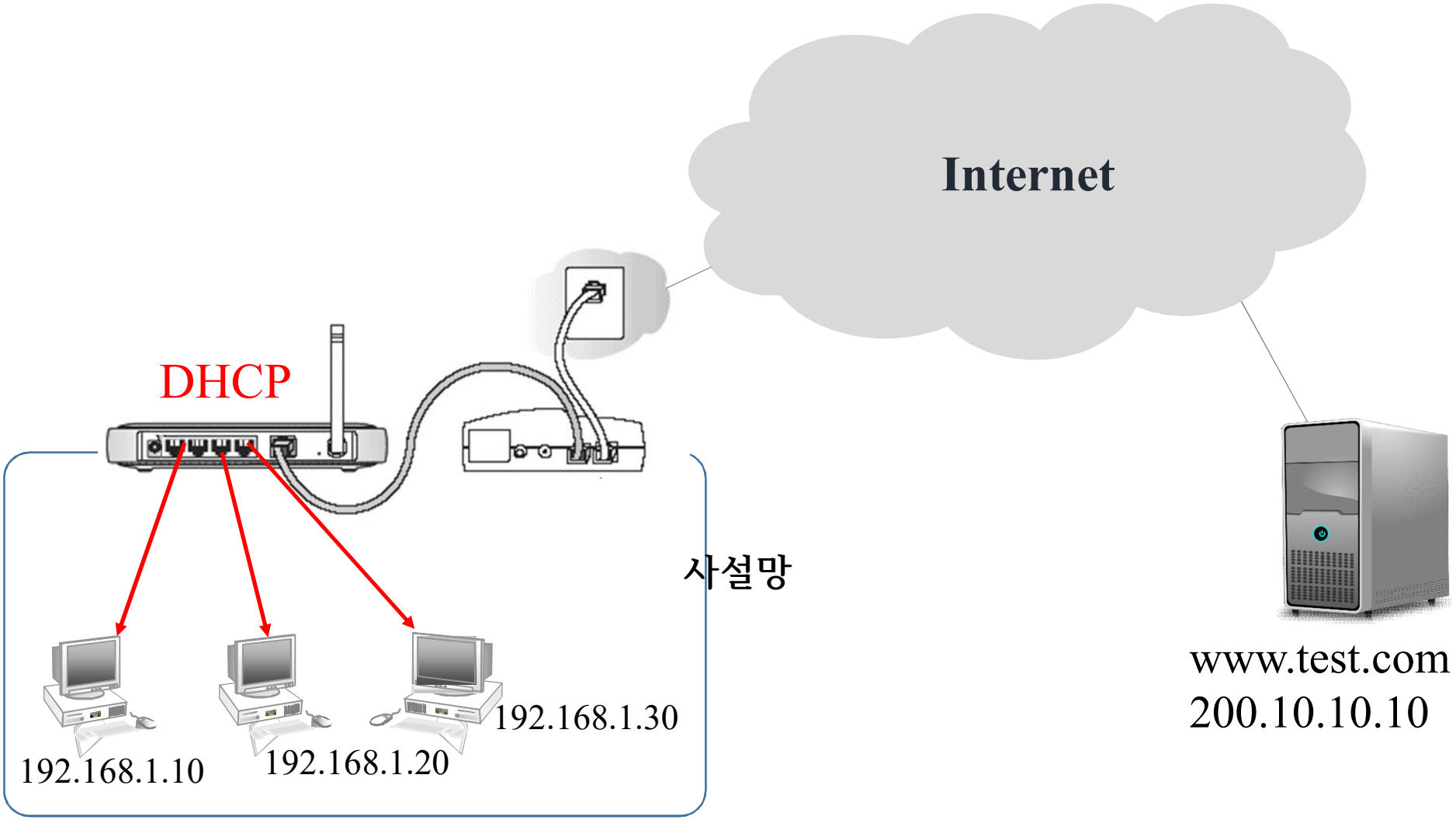
❶ **SNAT**

172.16.1.10	200.10.10.10

송신지 IP	수신지 IP
192.168.1.10	**200.10.10.10**

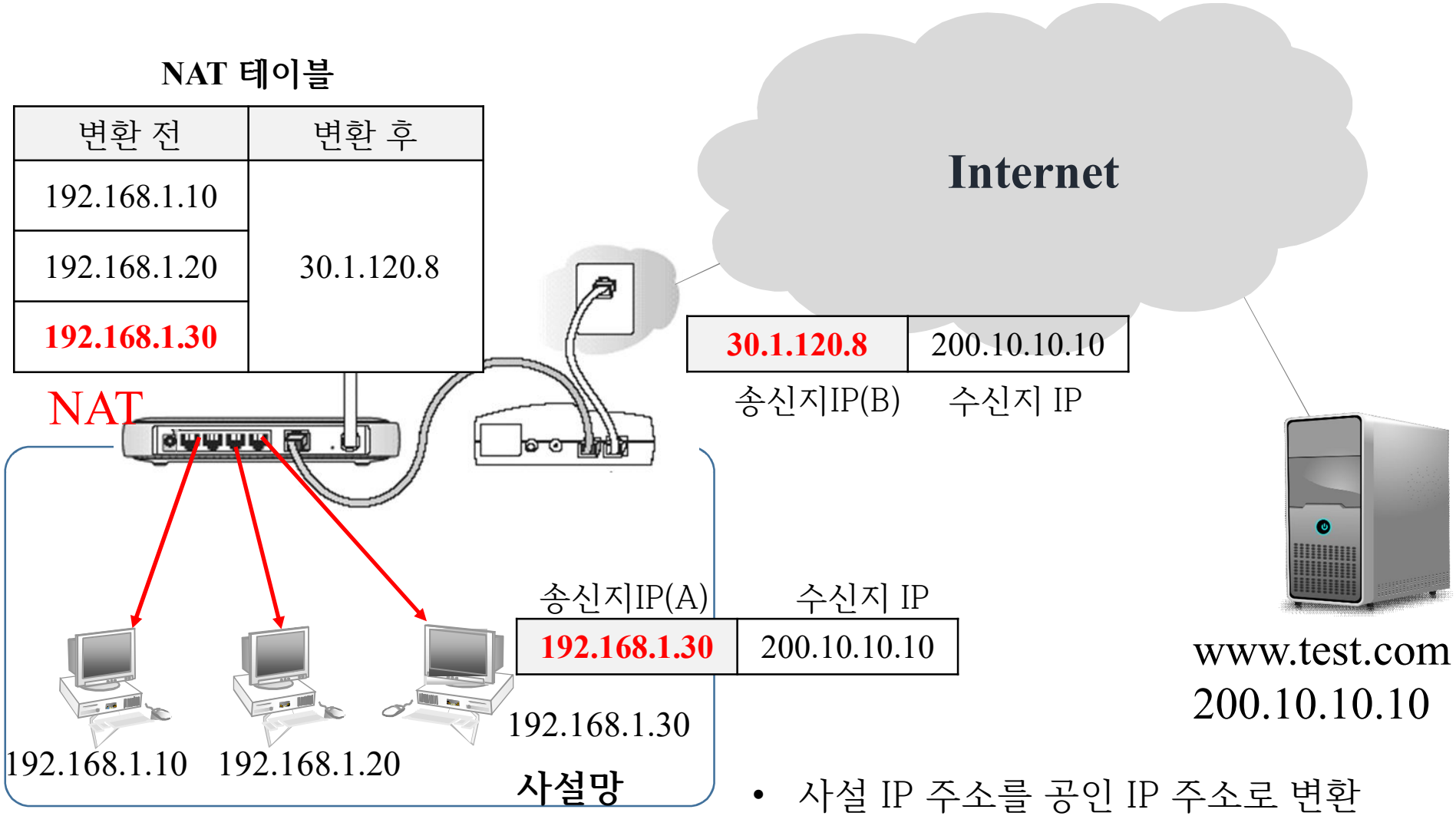
❷ **DNAT**

192.168.1.10	**10.10.10.10**


Internet
DHCP
사설망
192.168.1.10
192.168.1.20
192.168.1.30
www.test.com
200.10.10.10

NAT 테이블

변환 전	변환 후
192.168.1.10	30.1.120.8
192.168.1.20	
192.168.1.30	



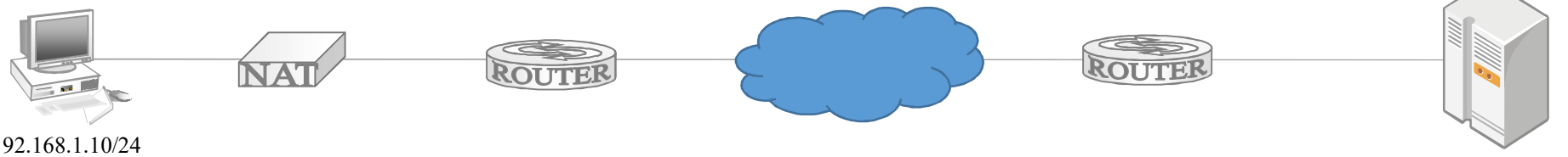
- 사설 IP 주소를 공인 IP 주소로 변환

Source NAT

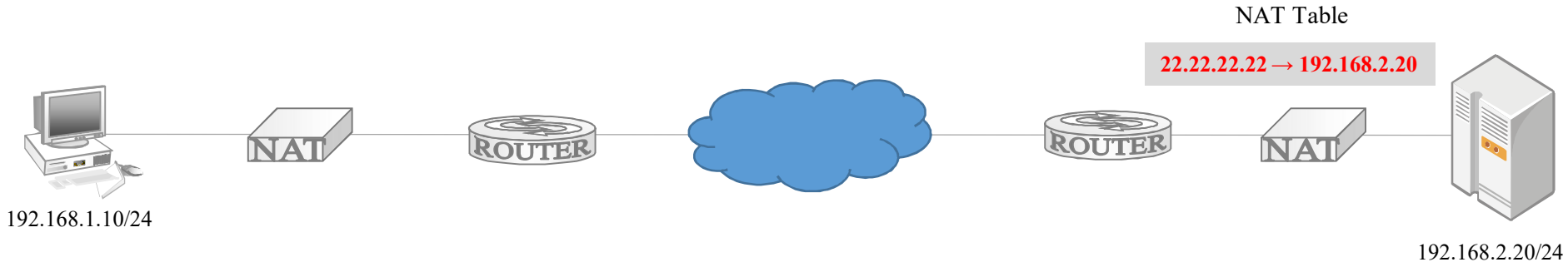
NAT Table

192.168.1.10 → 11.11.11.11

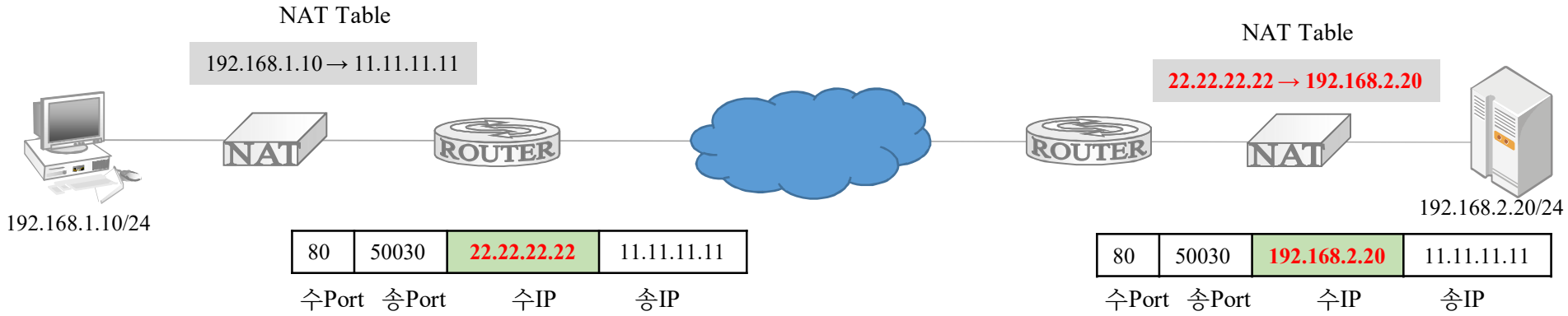
수Port	송Port	수IP	송IP
80	50030	22.22.22.22	11.11.11.11



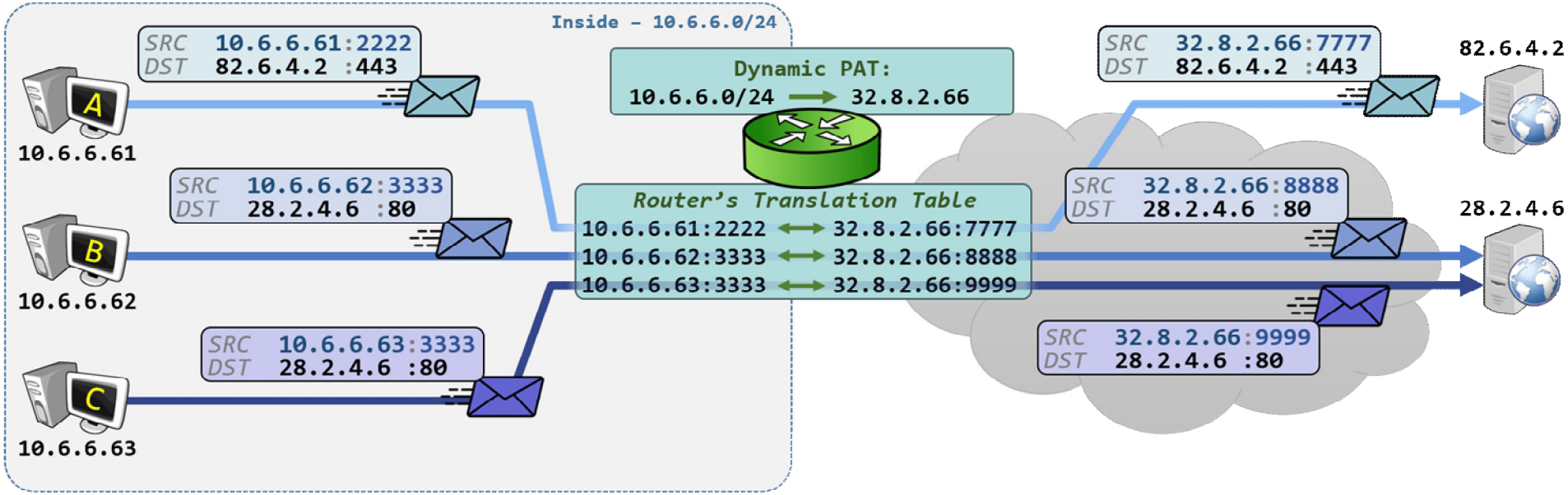
수Port	송Port	수IP	송IP
80	50030	22.22.22.22	192.168.1.10

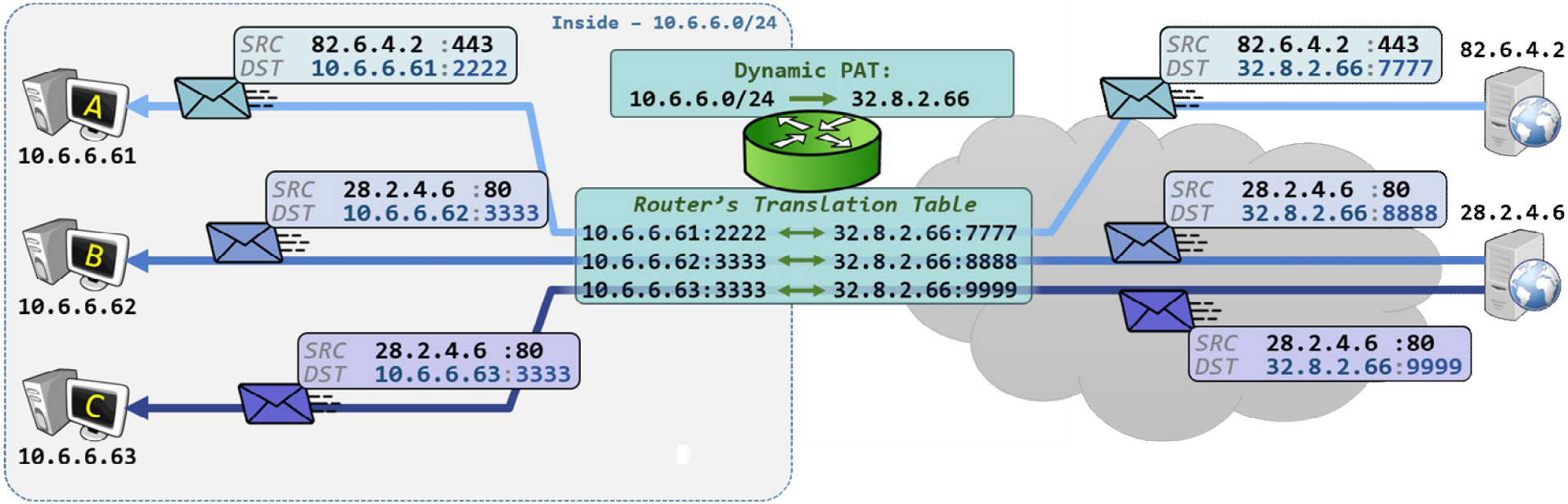


Destination NAT



PAT


Inside – 10.6.6.0/24
SRC 10.6.6.61:2222
DST 82.6.4.2 :443
A
10.6.6.61
SRC 10.6.6.62:3333
DST 28.2.4.6 :80
B
10.6.6.62
SRC 10.6.6.63:3333
DST 28.2.4.6 :80
C
10.6.6.63
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
Router's Translation Table
10.6.6.61:2222 ↔ 32.8.2.66:7777
10.6.6.62:3333 ↔ 32.8.2.66:8888
10.6.6.63:3333 ↔ 32.8.2.66:9999
SRC 32.8.2.66:7777
DST 82.6.4.2 :443
SRC 32.8.2.66:8888
DST 28.2.4.6 :80
SRC 32.8.2.66:9999
DST 28.2.4.6 :80
82.6.4.2
28.2.4.6


Inside – 10.6.6.0/24
SRC 82.6.4.2 :443
DST 10.6.6.61:2222
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
SRC 82.6.4.2 :443
DST 32.8.2.66:7777
82.6.4.2
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10.6.6.61
SRC 28.2.4.6 :80
DST 10.6.6.62:3333
Router's Translation Table
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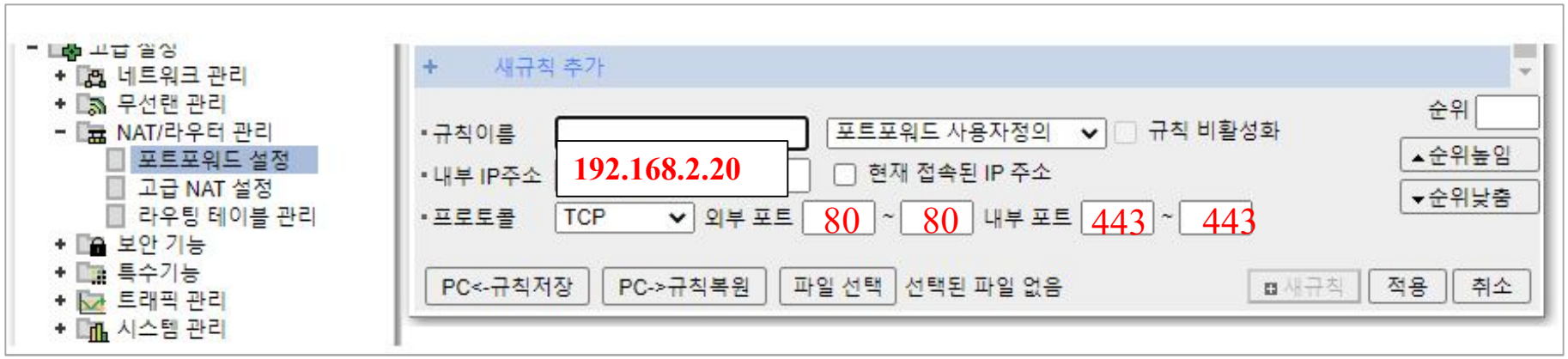
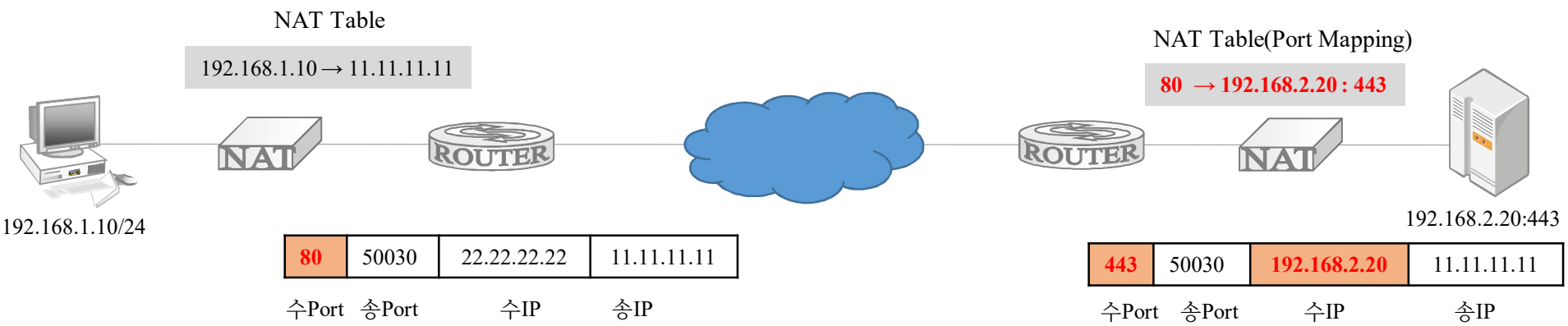
DST 32.8.2.66:8888
28.2.4.6
B
10.6.6.62
SRC 28.2.4.6 :80
DST 10.6.6.63:3333
SRC 28.2.4.6 :80
DST 32.8.2.66:9999
C
10.6.6.63

NAT Port Mapping (Port Forwarding)





http://200:1.1.10

HTTP

192.168.1.10:80

FTP

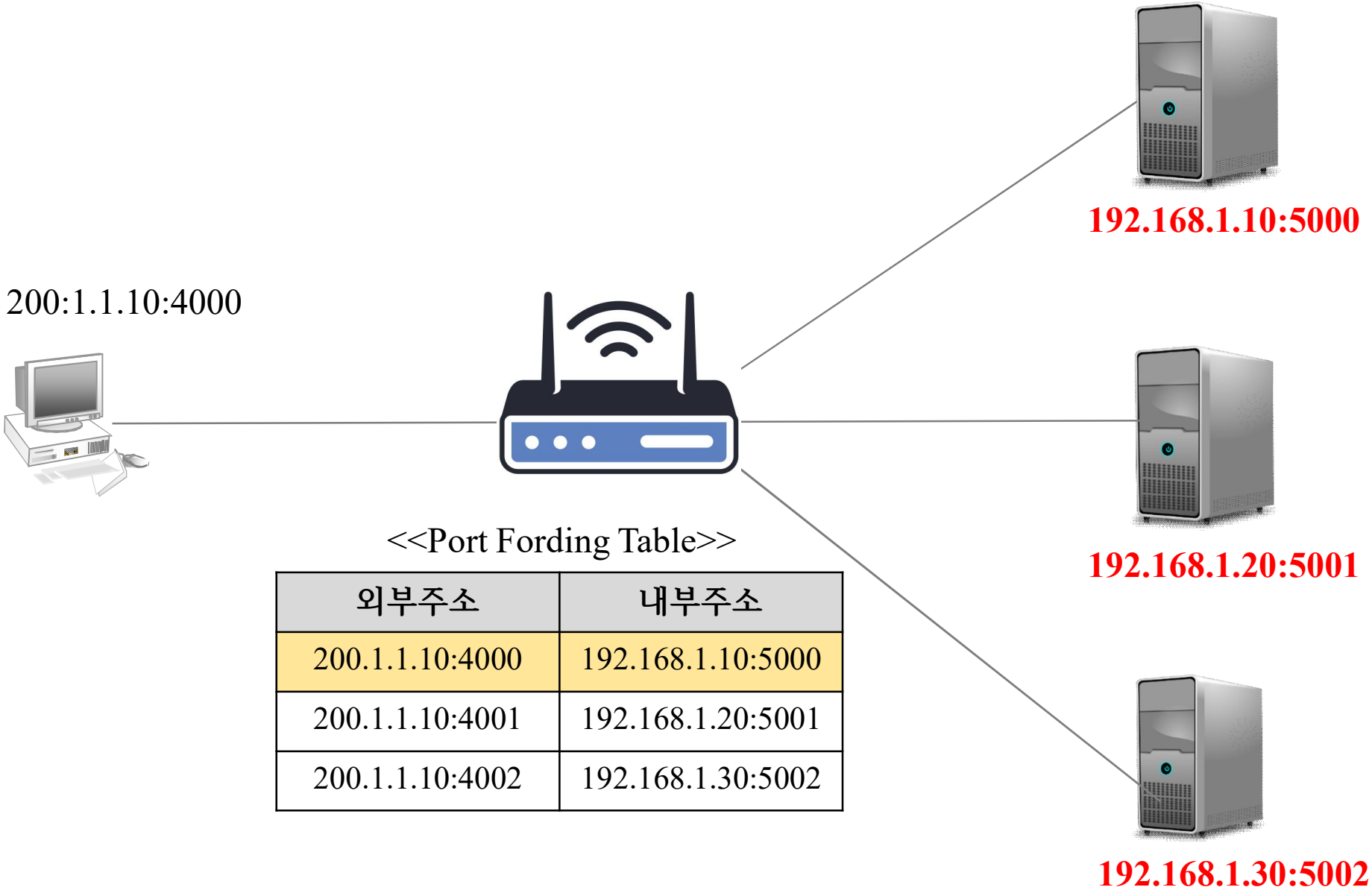
192.168.1.20:21

RDP

192.168.1.30:3389

<<Port Fording Table>>

외부주소	내부주소
200.1.1.10:80	192.168.1.10:80
200.1.1.10:21	192.168.1.20:21
200.1.1.10:3389	192.168.1.30:3389



<<Port Fording Table>>

외부주소	내부주소
200.1.1.10:4000	192.168.1.10:5000
200.1.1.10:4001	192.168.1.20:5001
200.1.1.10:4002	192.168.1.30:5002